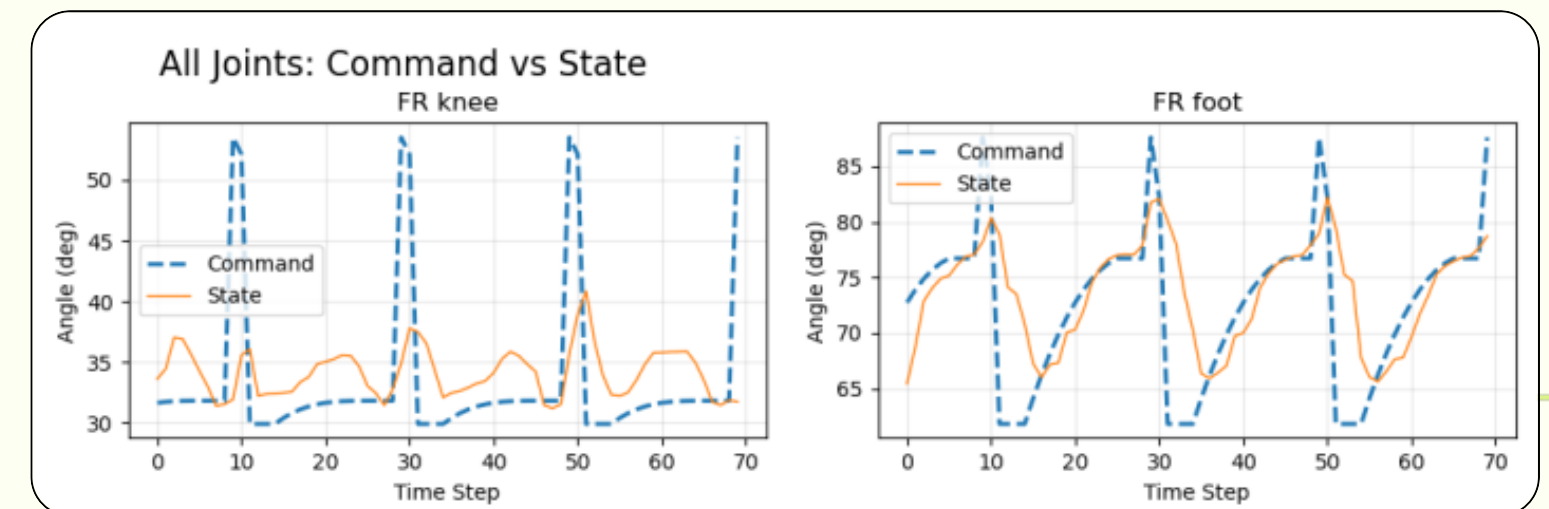
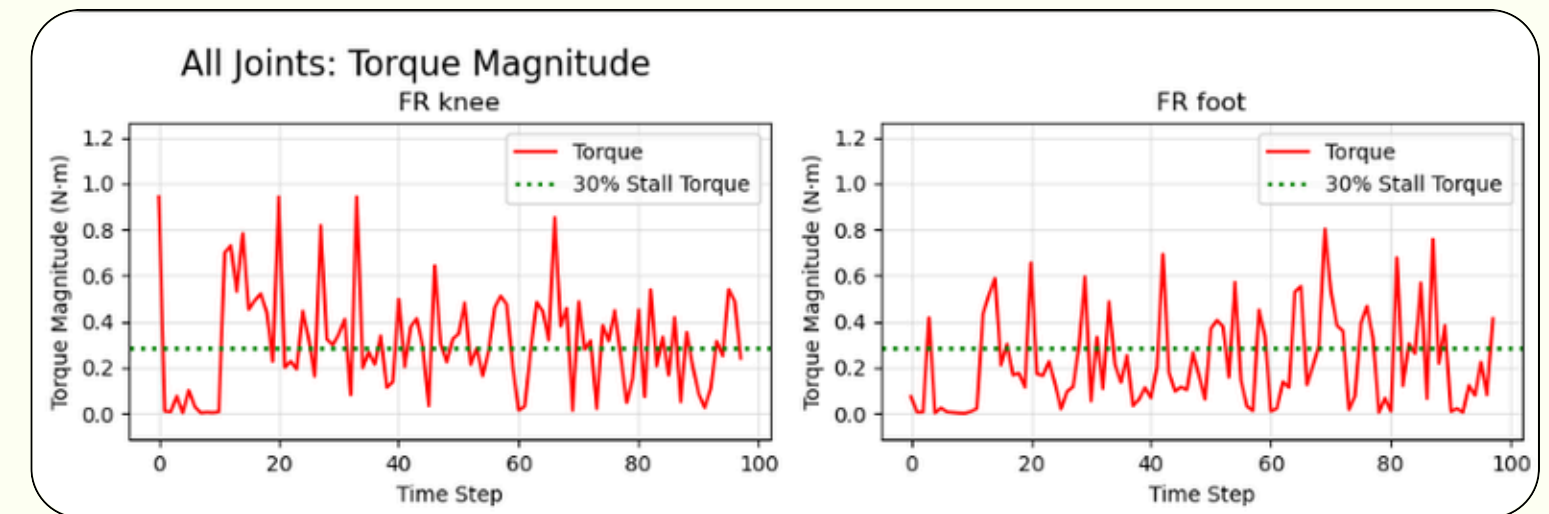


Simulation

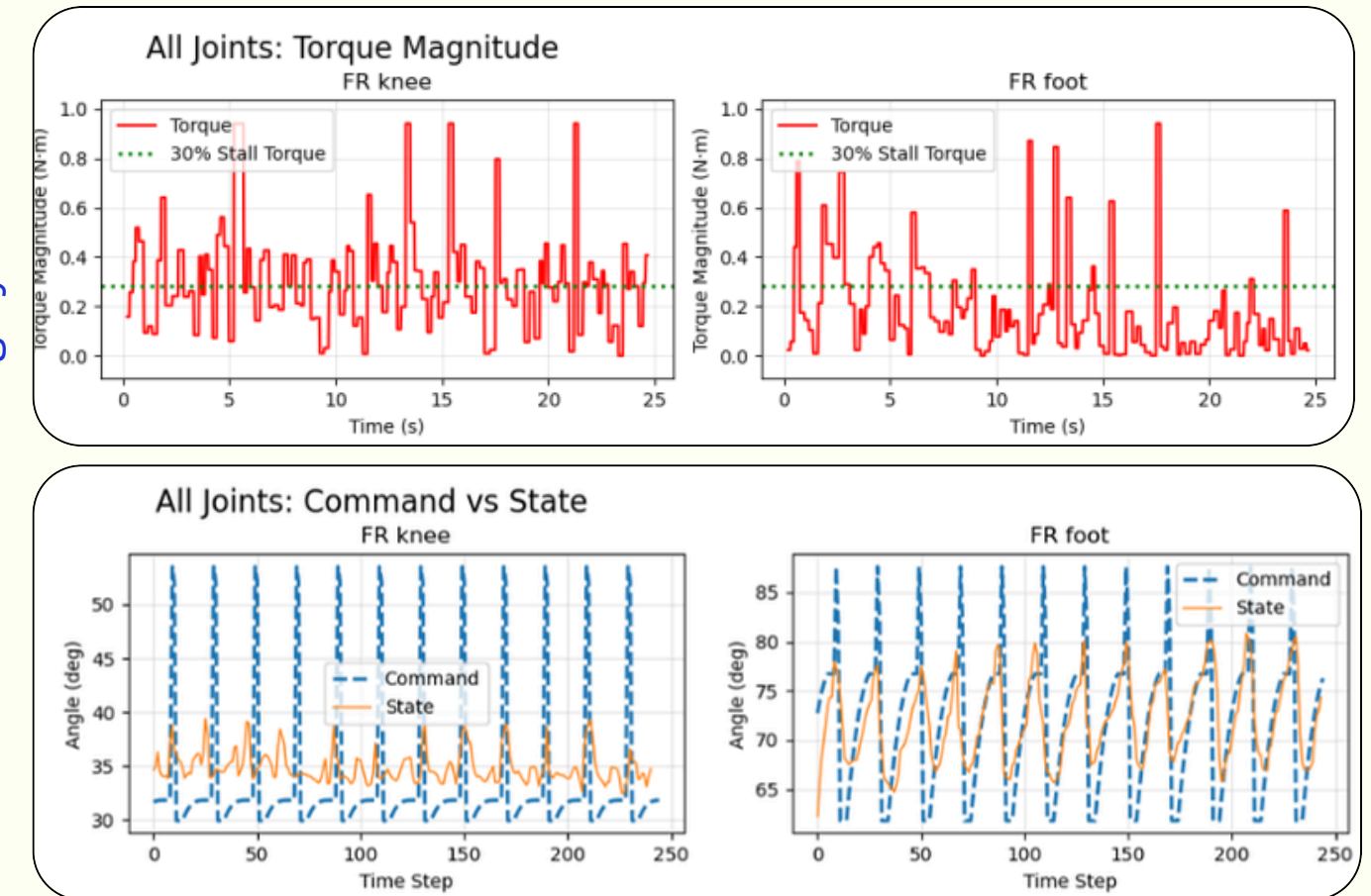
- Got the full sim pipeline running end-to-end: gait generation, kinematics, Gazebo integration
- Wired up force-torque sensors on all 12 joints to extract real torque profiles, the data needed to size the gravity-compensation springs.
- **Initial Results** : torque looked chaotic, with no visible gait structure and the actuator wasn't reaching its commanded position either.



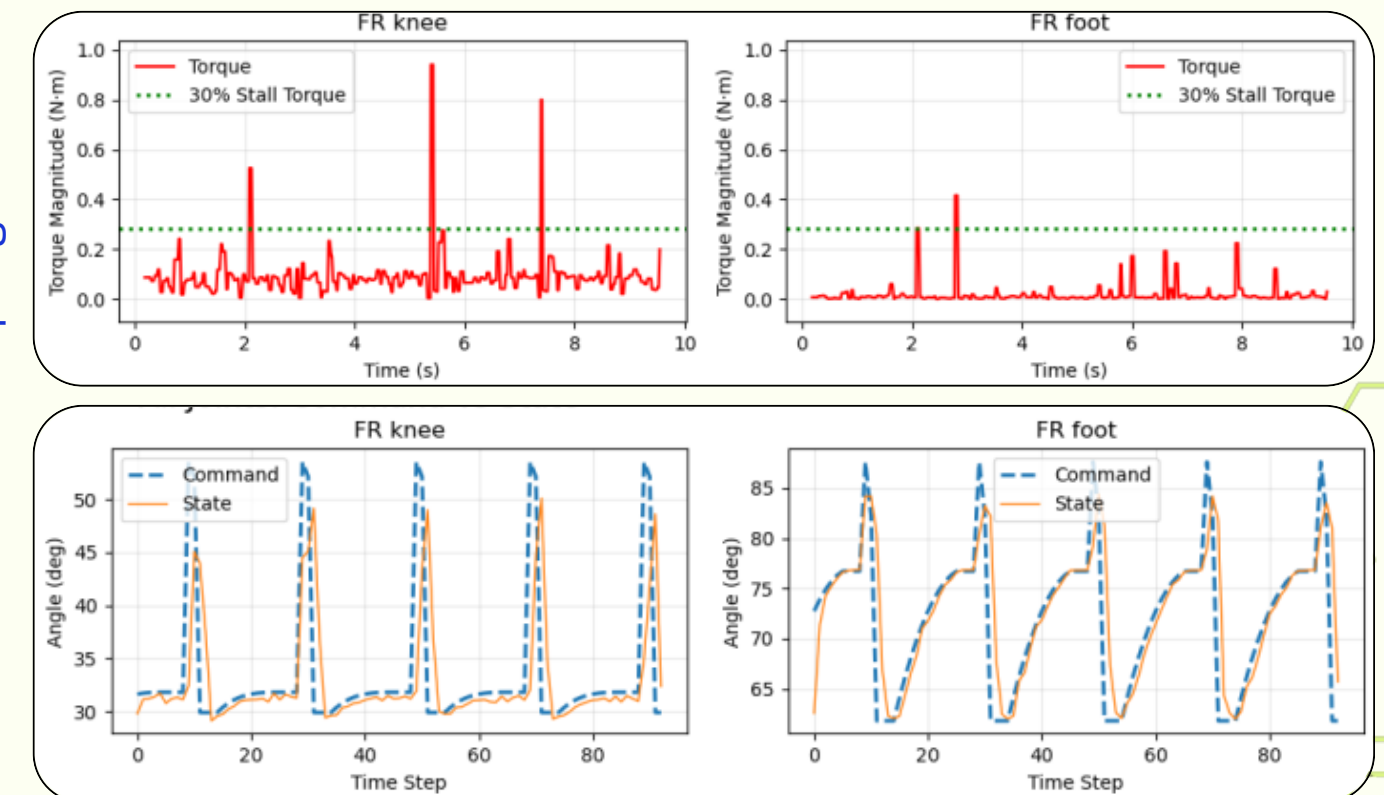
Chasing the Errors

- **First suspects :** Contact stiffness was left at Gazebo's high default, and identical PID gains were applied across hip, knee, and foot joints despite very different loading.
- Torque sampling was silently gated to 10Hz. The sensor genuinely publishes at 50Hz, but our logging was tied to the 10Hz gait loop. Decoupled it onto its own dedicated 50Hz timer.
- The real cause was sim time vs. wall-clock time. The simulation was actually running at only $\sim 0.11\times$ real-time, while our logging timer were paced by the wall clock so a 50Hz timer was re-reading stale data almost every tick.

Results after including dynamics

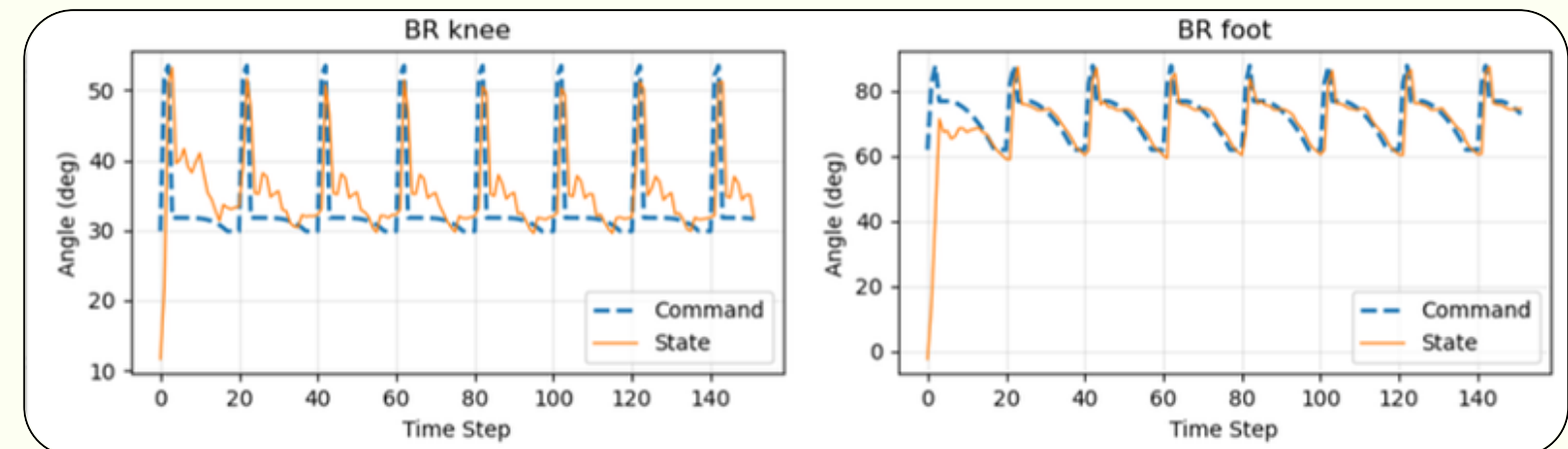
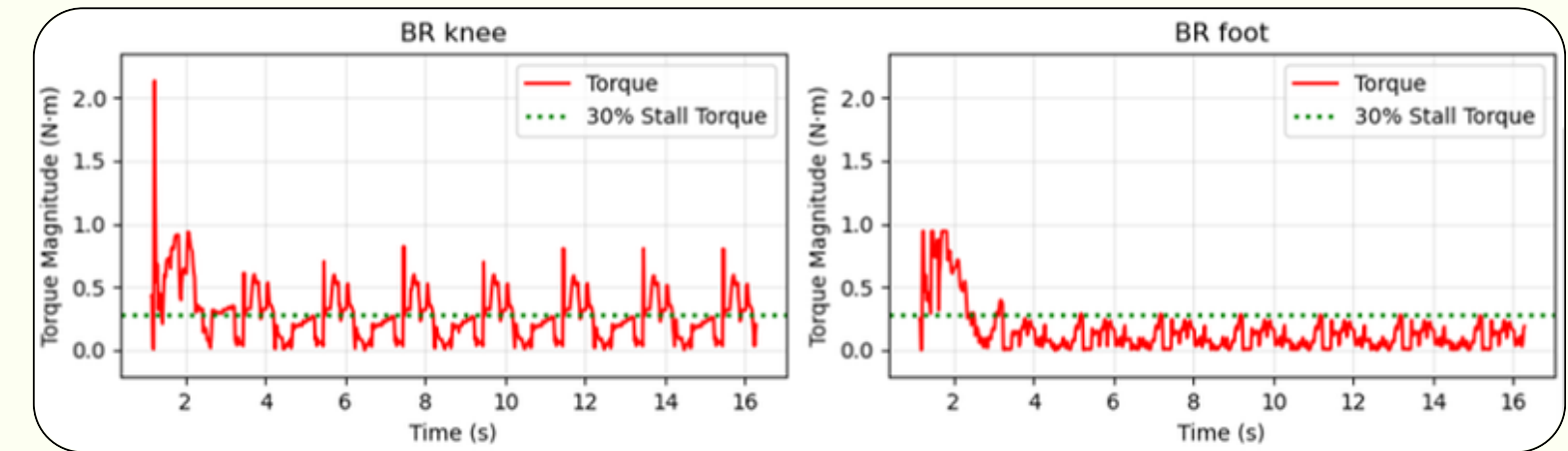


Results after decoupling timers

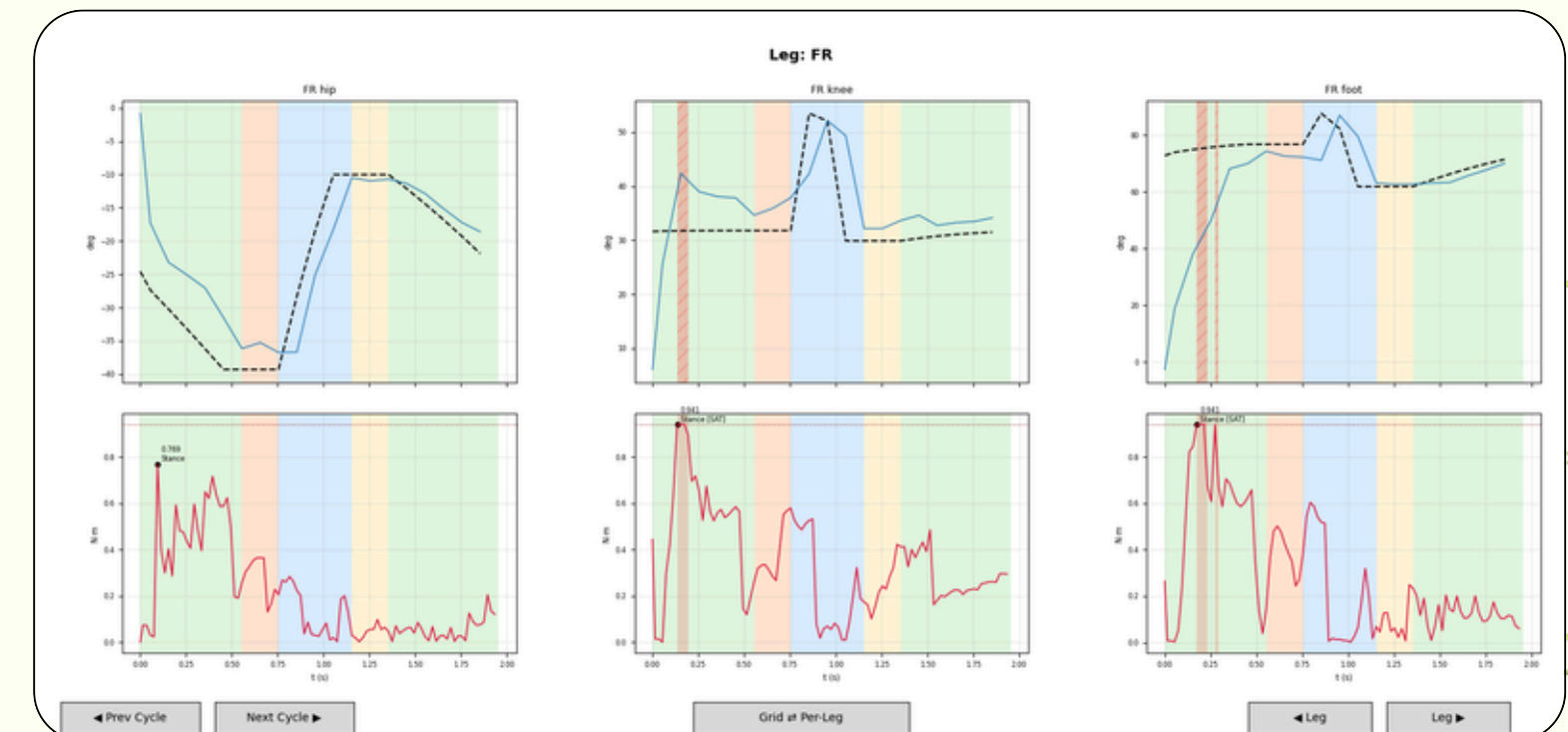


Fix and Results

- Switched the node to simulated time, so the gait loop and the torque logger both stay locked
- Result:** torque profiles went from chaotic to clean and periodic. the actuator now gets enough simulated time per step to actually converge.
- Also built an interactive gait-cycle viewer that shows phase shaded torque + position, per-cycle navigation, saturation highlighting



Accurate results and Interactive viewer





FUTURE PROSPECTS

- **Work on running the FEA analysis for 2π rotation with a Torque of 314.16 N·mm and verify stiffness and displacement values**
 - **Redesign the Spiral Torsion Spring to fit the frame of the robot, and run FEA Analysis for the redesigned spring**
 - **Workout a worm gear based mechanism to move the spring automatically to optimal positions based on the gait.**
 - **Figure out an optimal compliance ratio (λ)**
 - **3D print and test the springs and refine further from there.**
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